

Clément Vurpillot

Data Engineering | Machine Learning | MLOps
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SUMMARY

Final-year engineering student at ESILV, majoring in Data & Artificial Intelligence, available for a **6-month internship from February 2027** in Data Engineering, Machine Learning or MLOps. Hands-on experience across the full chain: data ingestion pipelines, machine learning, deep learning and LLM applications, REST APIs, containerised deployment and CI/CD on cloud services. Comfortable in Python, SQL and distributed data systems.

PROFESSIONAL EXPERIENCE

CERTAM

Rouen, France

Data & AI Engineering Intern (4 months)

Apr. 2026 – Aug. 2026

- Built an end-to-end IoT ingestion pipeline for 1 Hz vessel telemetry: J1939 CAN bus → AWS S3 (Parquet) → async Python → PostgreSQL/TimescaleDB.
- Cut dashboard load time from 3 hours to 30 seconds by moving computation into materialised views.
- Trained a Random Forest reconstructing particulate readings during sensor failure (R^2 up to 0.96), interpreted with SHAP.
- Deployed a Streamlit app and real-time Grafana alerting on a hardened Linux VPS; the pressure-drop alert caught a filter clogging before engine failure.

Diduenjoy

Paris, France

Data Labeling & Quality Supervisor (Internship, then Freelance)

July – Oct. 2024 | Aug. 2025

- Supervised a team of 5 interns under a contractual 72-hour SLA, with weekly quality audits.
- Labelled customer verbatims against business taxonomies and evaluated production NLP models (precision / recall).

PROJECTS

NLP analysis of 24k insurance reviews

Hugging Face, BERT, Llama-3, Sentence-Transformers

- Benchmarked a TF-IDF + logistic regression baseline (SHAP) against Sentence-Transformers embeddings + a PyTorch classifier for rating prediction; LDA topics, Word2Vec, semantic search.
- Built zero-shot topic detection (BART-Large-MNLI) and a **RAG** summarisation pipeline with cosine-similarity retrieval; switched from Flan-T5 to a locally deployed Llama-3 (8B) after error analysis showed hallucinations. Shipped as a Streamlit app.

CryptoIndexLab — containerised microservices platform

FastAPI, Docker, PostgreSQL, Nginx, AWS

- Team of 3: two FastAPI services (JWT auth, index management), 2 replicas each behind an Nginx reverse proxy and load balancer (my part), 3 isolated Docker networks, separate PostgreSQL databases.
- Hardened the CI/CD chain with multi-stage builds, non-root containers and a Trivy security scan gating on MEDIUM+ findings; demoed live on AWS Lightsail.

Full-stack MERN application with complete DevOps chain

Node.js, React, MongoDB, GitHub Actions

- Delivered in a 3-person Scrum team on Jira (PO / SM / Dev roles), from BRD / SRS / SDD and UML modelling to frontend, with unit and integration testing behind an **80% coverage gate** in CI/CD.

EDUCATION

ESILV – Graduate School of Engineering, Paris La Défense

Sept. 2022 – Aug. 2027

Engineering degree (Diplôme d'Ingénieur — CTI-accredited, EUR-ACE, Master's level, 300 ECTS), Data & Artificial Intelligence major. Coursework: Deep Learning (transformers, CNN/RNN), MLOps, Containerization, NoSQL, Information Retrieval & NLP, Data Mining, Explainable AI, Advanced Probability, Big Data (Spark), LLM & Generative AI.

RTU – Riga Technical University, Latvia

Jan. 2025 – June 2025

International Semester: data science, AI and engineering coursework, taught entirely in English.

TECHNICAL SKILLS

- Languages:** Python (Pandas, NumPy, Scikit-Learn), SQL, JavaScript / Node.js, Java, CQL.
- Machine Learning & AI:** TensorFlow, Hugging Face, transformers (BERT), LLMs and RAG, embeddings and vector search, deep learning, ensemble methods, NLP, data mining, statistical analysis, explainability (SHAP, LIME).
- Data & Distributed Computing:** PostgreSQL, TimescaleDB, IoT time-series, PySpark, MongoDB, Cassandra, Elasticsearch, Parquet, data modelling.
- Cloud & MLOps:** AWS (S3, Lightsail), GCP, Kubernetes (GKE), Docker, microservices, GitHub Actions, CI/CD, FastAPI / REST API development, Linux, Nginx.
- Engineering practice:** Git version control, automated testing, UML, Agile / Scrum, Jira, Grafana, Streamlit.

CERTIFICATIONS & LANGUAGES

Certifications: Scrum & Agile methods | IBM Data Fundamentals | IBM Getting Started with Data

Languages: French (native) | English C1 (TOEFL ITP 633/677) | German A2